

21shares

BASIS TRADING

A look into the cornerstone of institutional
crypto trading



WHAT IS BASIS TRADING?

It is the difference between the spot and future price of an asset:

As we get closer to expiry of the future the difference between them should converge to zero otherwise instant arbitrage can take place.

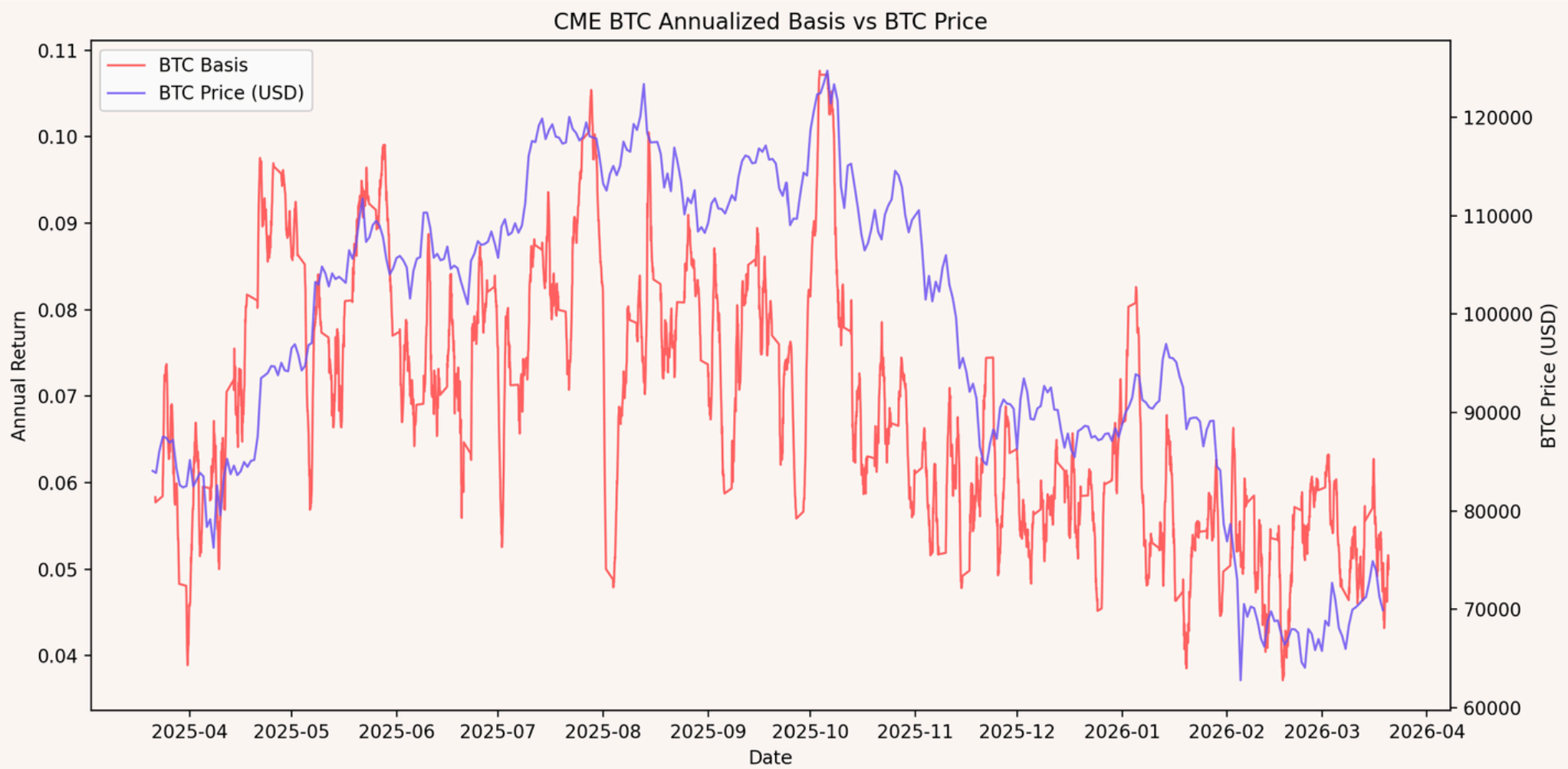
How did the basis trading concept come to be?

It all started back in 1979, when JF Eckstein identified a pricing inefficiency in Treasury futures, which traded at a discount to their underlying Treasuries. By buying futures and shorting the bonds, he could capture near risk-free profits as the basis-the price difference-converged.

However, price divergence required more collateral, which forced Eckstein to sell his position to Salomon Brothers, where John Meriwether (see: When Genius Failed) took over and turned the strategy into a lucrative venture, becoming partner soon after.

By the 1990s, basis trading had evolved, becoming an industry staple with futures contracts generally priced at a premium instead of a discount.

The trade started in the Fixed Income world but has evolved to be incorporated across asset classes today.

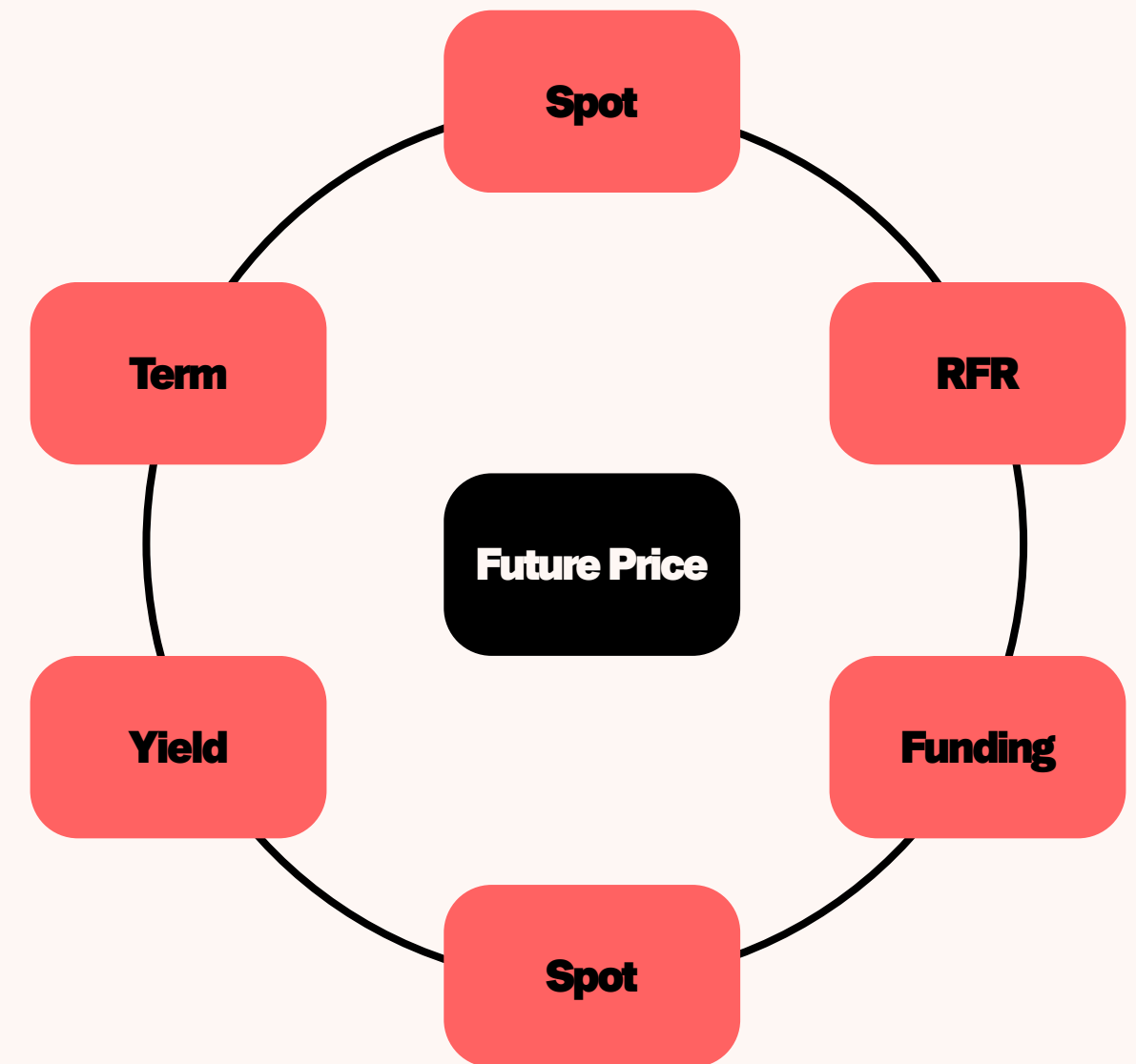


Sources: CME, 21Shares

WHAT DRIVES THE FUTURES PRICE?

The fair value of a future is determined by a number of factors:

- **Spot:** fundamentally linked to futures, given it's the current market price of an asset.
- **Risk-Free Rate (RFR):** interest rate on a theoretically risk-free investment. Higher RFR increases the futures price, as holding cash and earning interest becomes more attractive.
- **Funding Rate:** cost of borrow above RFR, where the costlier it is to finance positions the higher the futures price are going to be.
- **Repo:** rate at which financial institutions borrow or lend securities for short-term financing. The higher it is, the lower the futures prices are going to be to stay attractive.
- **Yield:** return generated by the underlying (i.e., dividends in equities). Futures may adjust downward to reflect the benefit of holding the asset instead of the future.
- **Term:** time to contract expiration. Longer-term futures contracts can be affected by storage costs, interest rates, volatility expectations and market liquidity.



*Additional factors that can also influence FV of futures include: WHT, CTD, storage costs etc.

WHY DO BASIS TRADE OPPORTUNITIES EXIST?

Fair value represents the theoretically correct price of a futures contract, derived from the 6 elements mentioned earlier.

So, does everyone trade with the same fair value in mind? - No!

The fair value framework is not universal, and behavioural biases can create extra distortions:

1. Fair value is investor-specific:

- Different **repo** rates mean investors have varying costs to finance positions.
- **Staking** yields show discrepancies depending on provider.
- Cost of borrow is expressed as **RFR** and **funding rate**, which is not fixed.

2. Behavioural concerns:

- Market participants may prefer futures over spot due to **lower capital requirements** or leverage, pushing futures above fair value.
- Investors follow **momentum**, amplifying deviations from fair value.
- Everyone participates in the market with different **expectations** in mind.
- Many crypto investors are **strong believers** in the long-term potential of the asset class, so they are willing to pay a premium on futures contracts, expecting price appreciation, regardless of short-term valuation models.

This means that what is mispricing for one trader may be fair value for another.

OPTIMIZE WITH STAKING!

Staking yield reshapes basis dynamics. Basis trading can be executed via spot or ETPs, but using staking-enabled ETPs as the spot leg offers a clear edge. By embedding staking yield, the fair value of the basis moves lower than with spot alone (assuming yield > management fees), **boosting investor returns** for the same market-traded basis.

Staking coins directly requires managing staking providers, navigating lockup periods, and handling custody risks. ETPs eliminate these hurdles and are thus more readily accepted by **Prime Brokers as collateral** for financing the futures leg. By pledging ETPs instead of selling underlying securities, investors can maintain exposure to their holdings while accessing liquidity for new positions, reducing forced liquidations during volatile market conditions.

PnL = Mkt value of Basis - FV of the Basis.

ETH BASIS TRADE WITHOUT STAKING					
Spot	1750	FV Future	1781	MV Future	1794
RFR	4.5%	FV Basis	6.99%	MV Basis	10.06%
Funding	2%	Diff Basis	3.07%		
Repo	1%				
Staking	-				
Term	25%				
MF	1.49%				

ETH BASIS TRADE WITH STAKING					
Spot	1750	FV Future	1767	MV Future	1794
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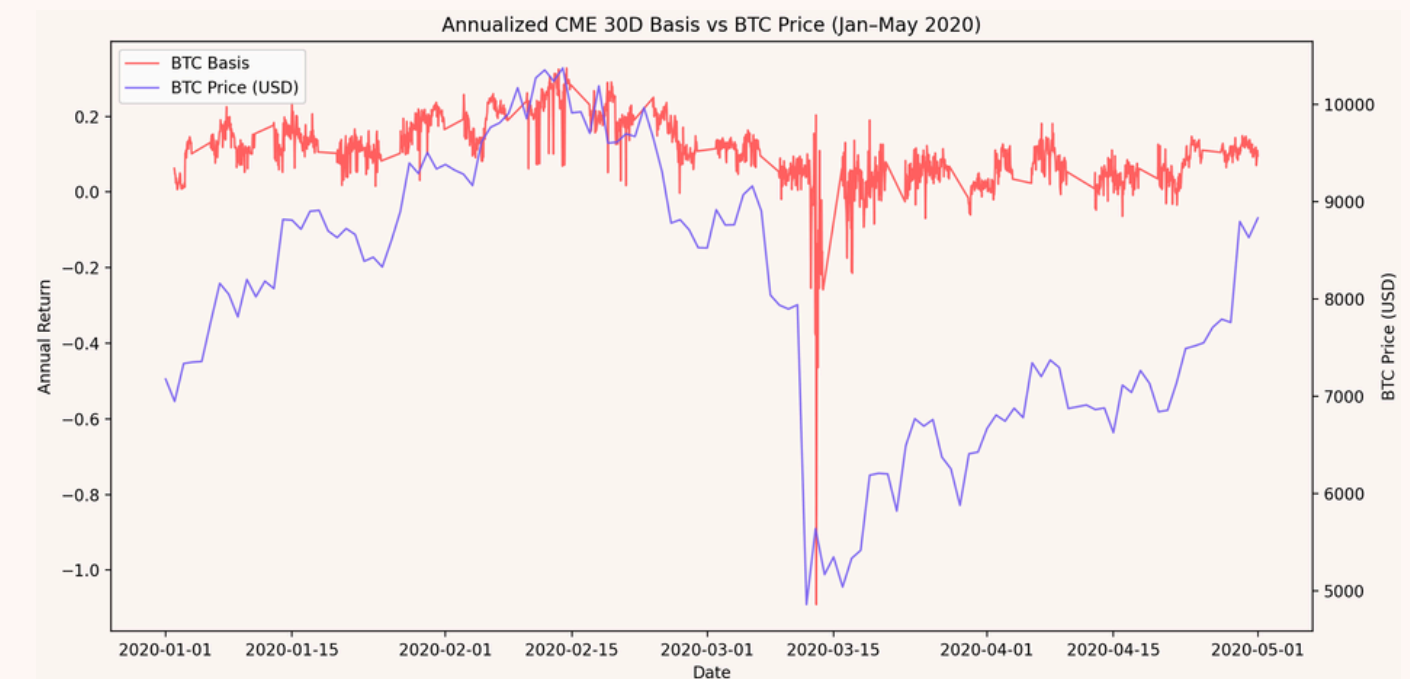
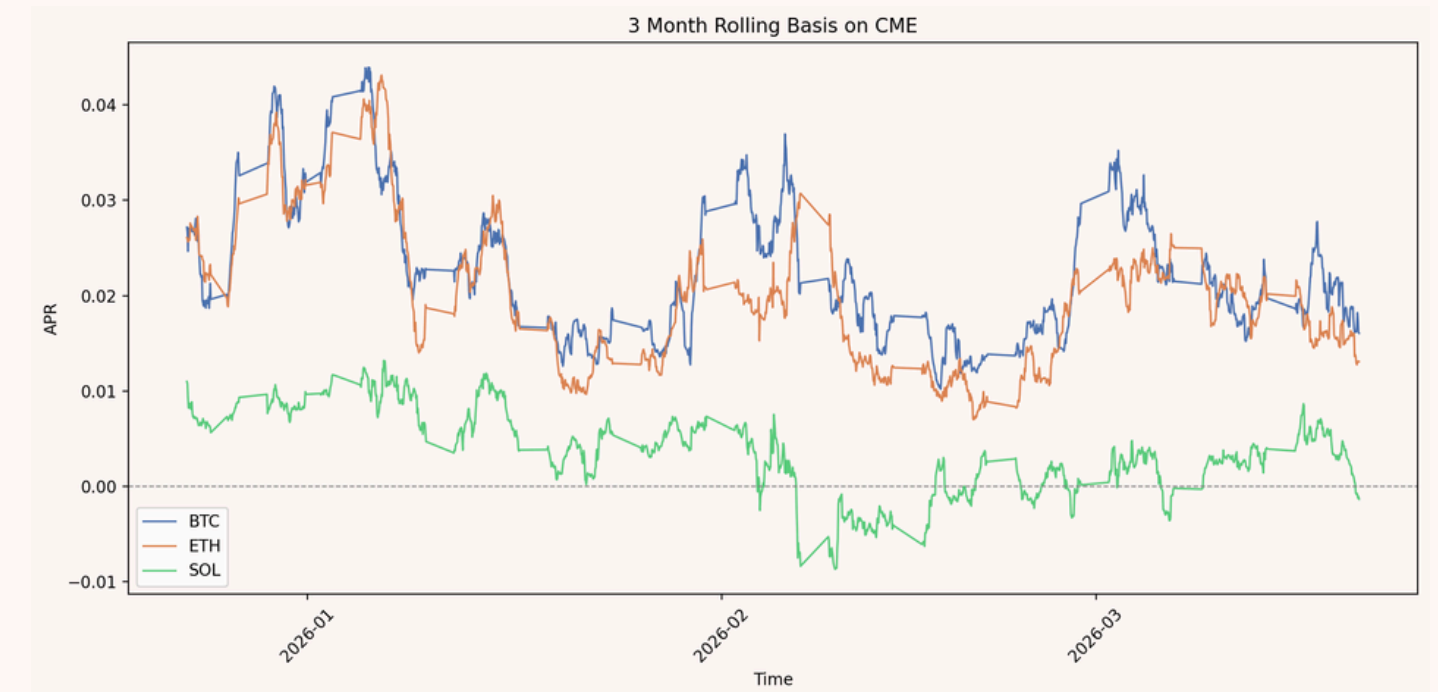
HOW TO IDENTIFY A BASIS TRADE OPPORTUNITY?

Depends on the coin:

- Bitcoin cannot be staked, and has the highest popular demand, pushing the future market price higher than its fair value.
- Ethereum can be staked, with average reward rate ~3%, approximately offsetting management fees, meaning the basis market value is lower than Bitcoin, more due to Bitcoin's popularity, all else equal.
- Solana can also be staked, but with higher rewards at ~8%, thus showcasing the lowest basis market value of the three.

Example: Covid-19

An aggressive “dash-for-cash” situation like March 2020, showcased the benefits of trading basis correctly, with consideration of entry points. As Bitcoin's price plummeted nearly 40% in a single day, traders holding leveraged positions faced margin calls and those unable to meet collateral requirements were forced to liquidate. This led to futures trading substantially lower against spot, creating significant positive returns for basis traders.



HOW TO TRADE BASIS? - PB & MM

Prime Brokers (PBs) play a key role in basis trading by providing the infrastructure and capital efficiency needed to execute and manage trades effectively.

PBs facilitate **simultaneous execution** of both the spot and futures legs, ensuring liquidity and reducing slippage. They also offer financing solutions that optimize capital deployment.

Access to a Prime Broker is crucial for institutional investors, as it enables more efficient trade execution, margin management, and risk mitigation when engaging in basis trading.

Key Functions of Prime Brokers in Basis Trading:

- Provide leverage/financing to enhance capital efficiency
- Facilitate seamless trade execution across spot and futures markets.
- Offer collateral solutions, with ETPs often being more accepted than direct crypto holdings.
- Mitigate counterparty risk, ensuring secure settlement.

ETPs with strong liquidity and support from market makers allow investors to benefit from the tightest spreads of the market.

INSTITUTIONAL PRIME BROKERS FOR BASIS TRADING

Cantor Fitzgerald L.P.



Barclays



Goldman Sachs



Bank of America Merrill Lynch



INSTITUTIONAL MARKET MAKERS FOR BASIS TRADING

DRW Europe B.V.



Flow Traders B.V.



Goldenberg Hehmeyer LLP



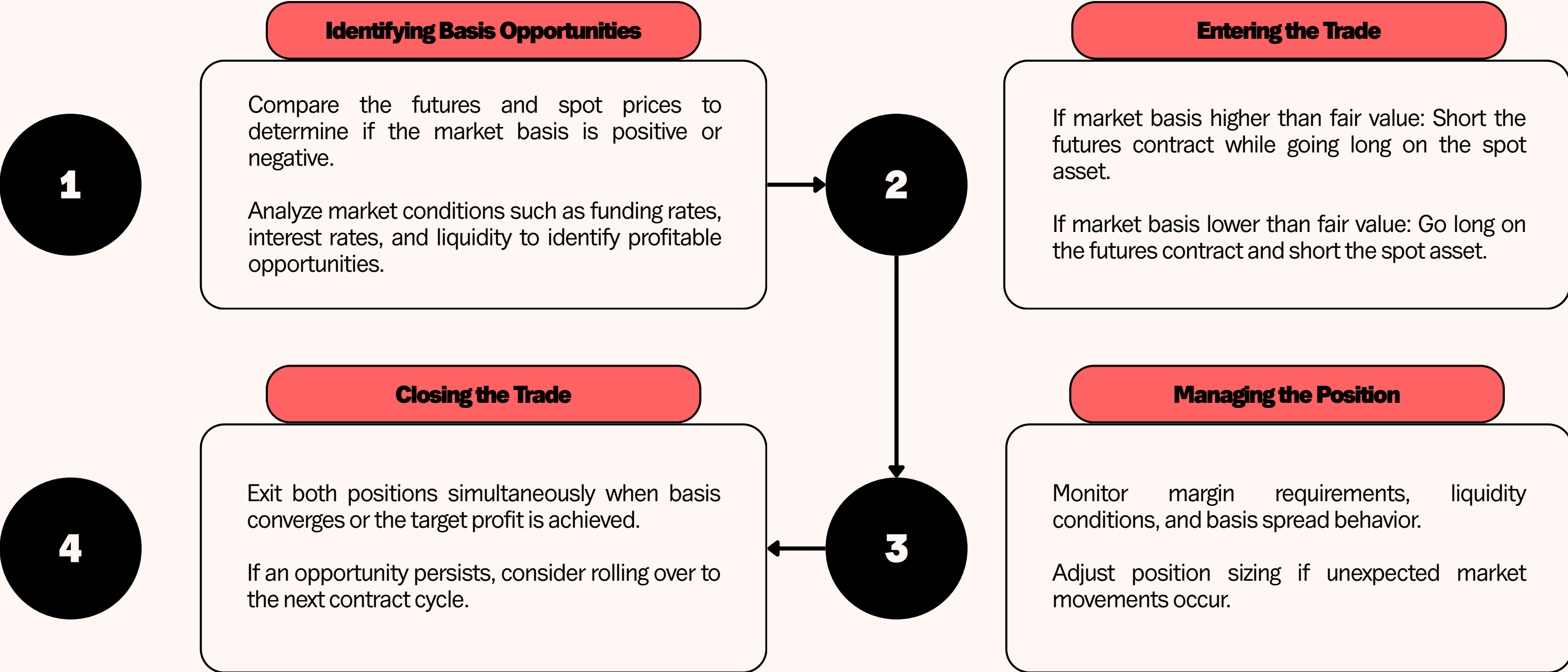
Jane Street Financial Limited



Virtu Financial Ireland Limited



HOW TO TRADE BASIS? - TRADE LIFE CYCLE



HOW TO TRADE BASIS? - WHAT TO CONSIDER

Trade execution is crucial to maximizing basis trade profitability.

Monitoring liquidity distribution throughout the day or trading OTC , might help ensuring execution at peak liquidity, minimizing slippage and optimizing capital efficiency. Investors might also want to consider market depth, order book imbalances, and trading session overlaps to further improve the execution.

CONSIDER	IMPACT ON EXECUTION	BEST PRACTICE
Market Depth	Deeper liquidity allows better order execution.	Trade ETPs and CME futures contracts with high volume. Trading OTC may offer better fills for large trade. Work with multiple OTC desks might help securing competitive pricing.
Slippage	Large orders can move markets, reducing arbitrage profits.	Execution strategies such as split the orders, OTC trading, VWAP or TWAP can help minimize the impact. Work with market makers and Prime Brokers to execute large trades, and consider trading off-book.
Margin Calls	If margin account falls below required level, positions may be liquidated if no additional funds are provided.	Regularly check margin requirements and maintain a buffer to avoid unexpected calls.
Trading Hours	Crypto trades 24/7, but ETPs follow traditional market hours.	Execute during overlapping hours to maximize liquidity. Trading OTC might allow to execute outside market hours, possibly providing a wider window of overlap.

HOW TO TRADE BASIS? - RISK MANAGEMENT

Basis trading is delta-neutral but still exposes investors to funding rate volatility, liquidity risks, and counterparty risks, which may impact trade profitability.

Implementing effective risk management and continuous monitoring might help achieve optimal returns and capital protection.

RISK	IMPACT ON BASIS TRADE	MITIGATION STRATEGY
Financing Rates Volatility	Changes in financing rates for hedging instruments can compress arbitrage spreads and impact profitability.	Monitor financing rates from prime brokers. Adjust hedges or consider alternative hedging instruments if costs rise.
Liquidity	ETPs, futures, and hedging instruments may have variable market depth, affecting execution efficiency. Poor liquidity can widen spreads and increase trading costs.	Use ETPs and futures with deep liquidity. Implement execution strategies to minimize price impact. Work with market makers and prime brokers to access off-exchange liquidity.
Counterparties	Prime brokers and custodians may introduce counterparty risks.	Work with regulated prime brokers and ETPs with transparent custodianship.
Rollover	Futures and swaps used for hedging require periodic rollovers, which can introduce additional costs.	Plan rollovers ahead of expiry and use spread trading strategies to mitigate impact. Diversify hedging instruments to reduce dependency on a single product.



APPENDIX

FAIR VALUE OF FUTURE FORMULA*

$$\text{FV Future} = \text{Spot} \times [1 + (\text{RFR} + \text{Funding} - \text{Repo} - \text{Staking} + \text{MF}) \times \text{Term}]$$

$$\text{FV Basis} = [(\text{FV Future} \div \text{Spot}) - 1] \div \text{Term}$$

$$\text{Diff Basis} = \text{MV Basis} - \text{FV Basis}$$

*Simplified for demonstration purposes

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